
University Name

Department of Computer Science & Engineering

Thesis Title Goes Here

Submitted in partial fulfillment of the requirements for the degree of

**Bachelor of Science in Computer Science &
Engineering**

Submitted by

Your Full Name

Student ID: STUDENT-ID

Supervised by

Supervisor Name

Month Year

Declaration

I hereby declare that the work presented in this thesis titled “**Thesis Title Goes Here**” is my own original work carried out as an undergraduate student at **University Name**, under the supervision of **Supervisor Name**.

All information in this thesis has been obtained and presented in accordance with academic rules and ethical conduct. I have fully cited and referenced all material and results that are not original to this work in accordance with the standard referencing norms.

I also declare that this thesis has not been submitted elsewhere for the award of any other degree or diploma.

Signature of the Author

Date

Your Full Name

STUDENT-ID

Department of Computer Science & Engineering

University Name

Approval

This thesis titled “**Thesis Title Goes Here**”, submitted by **Your Full Name** (Student ID: STUDENT-ID) to the **Department of Computer Science & Engineering, University Name**, has been accepted as satisfactory in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science & Engineering**.

Board of Examiners:

Supervisor

Date

Supervisor Name

Department of Computer Science &
Engineering

University Name

Head of Department

Date

Name of HoD

Department of Computer Science &
Engineering

University Name

External Examiner

Date

Name of Examiner

Institution Name

Abstract

Title : Thesis Title Goes Here
Author : Your Full Name
Student ID : STUDENT-ID
Degree : Bachelor of Science in Computer Science & Engineering
Supervisor : Supervisor Name
Institution : Department of Computer Science & Engineering, University Name
Date : Month Year

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Keywords: keyword one, keyword two, keyword three, keyword four, keyword

five

Acknowledgements

I would like to express my deepest gratitude to **Supervisor Name** for their invaluable guidance, continuous encouragement, and insightful feedback throughout the course of this research. Their expertise and dedication have been instrumental in shaping this work.

I extend my sincere thanks to the faculty members of the Department of Computer Science & Engineering for their support and for providing the academic environment necessary to complete this project.

I am grateful to my colleagues and lab partners for the fruitful discussions, collaborative spirit, and moral support during the challenging phases of this research.

Finally, I owe an immeasurable debt of gratitude to my family for their unwavering love, patience, and encouragement throughout my academic journey.

Your Full Name
University Name
Month Year

Contents

Declaration	i
Approval	ii
Abstract	iv
Acknowledgements	vi
List of Figures	x
List of Tables	xi
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Research Objectives	1
1.4 Research Questions	2
1.5 Scope and Limitations	2
1.6 Thesis Organization	2
2 Literature Review	3
2.1 Traditional Methods	3
2.2 Deep Learning Approaches	3
2.2.1 Recurrent Neural Networks	4
2.2.2 Convolutional Approaches	4
2.3 Transformer-Based Models	4
2.4 Comparative Summary	4
2.5 Research Gap	4
3 Problem Statement and Dataset	6
3.1 Formal Problem Definition	6
3.2 Dataset Description	6
3.2.1 Source and Collection	6
3.2.2 Dataset Statistics	6
3.2.3 Exploratory Data Analysis	7
3.3 Preprocessing Pipeline	7

4	Methodology	8
4.1	Overview of Proposed Approach	8
4.2	Model Architecture	8
4.2.1	Embedding Layer	8
4.2.2	Recurrent Layer	8
4.2.3	Bidirectional Configuration	9
4.2.4	Classification Head	9
4.3	Training Configuration	9
4.3.1	Loss Function	9
4.3.2	Optimization	10
4.3.3	Hyperparameter Summary	10
5	Implementation	11
5.1	Development Environment	11
5.2	Code Structure	11
5.3	Key Implementation Details	12
5.3.1	Custom Dataset Class	12
5.3.2	Model Definition	12
5.3.3	Training Loop	13
5.3.4	Reproducibility	13
6	Results	15
6.1	Overall Performance Summary	15
6.2	Training and Validation Curves	15
6.3	Confusion Matrix Analysis	15
6.4	Per-Class Performance	15
6.5	Ablation Study	17
6.5.1	Effect of Bidirectionality	17
6.5.2	Effect of Number of Layers	17
6.5.3	Effect of Dropout Rate	17
6.5.4	Optimizer Comparison: Adam vs. SGD	17
7	Discussion	19
7.1	Interpretation of Results	19
7.2	Comparison with Related Work	19
7.3	Limitations	19
7.4	Ethical Considerations	20
8	Conclusion and Future Work	21
8.1	Conclusion	21

8.2 Future Work	21
References	23

List of Figures

3.1	Class distribution of the dataset	7
4.1	Overview of the proposed system pipeline	8
6.1	Training and validation accuracy/loss — Best Bi-GRU model	16
6.2	Normalized confusion matrix of the best-performing Bi-GRU model (72.8%)	16

List of Tables

2.1	Comparison of related works	5
3.1	Dataset class distribution	6
3.2	Preprocessing configuration summary	7
4.1	Hyperparameter configurations used in experiments	10
5.1	Development and training environment	11
6.1	Test accuracy across all model configurations	15
6.2	Per-class precision, recall, and F1-score — Best Bi-GRU model . . .	16

Chapter 1

Introduction

1.1 Background and Motivation

Provide a broad overview of the research domain here. Introduce the general topic and establish context for the reader. Explain why the problem is important and relevant.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1.2 Problem Statement

Clearly define the specific problem this thesis addresses. State the gap in existing knowledge or limitations of prior work.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

1.3 Research Objectives

The main objectives of this research are:

1. To investigate ...
2. To design and implement ...
3. To evaluate ...
4. To compare ...

1.4 Research Questions

This thesis aims to answer the following research questions:

RQ1. How does ...?

RQ2. What is the effect of ...?

RQ3. Can ... achieve ...?

1.5 Scope and Limitations

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

1.6 Thesis Organization

The remainder of this thesis is organized as follows:

Chapter 2 provides a comprehensive review of related literature and state-of-the-art methods.

Chapter 3 formally defines the problem and outlines the research methodology.

Chapter 4 describes the proposed approach, model architecture, and experimental setup.

Chapter 5 covers implementation details, tools, and datasets used.

Chapter 6 presents experimental results and quantitative evaluation.

Chapter 7 discusses findings, insights, and comparisons with prior work.

Chapter 8 concludes the thesis and outlines directions for future work.

Chapter 2

Literature Review

This chapter provides a systematic review of the existing literature relevant to ... The review is organized thematically to highlight key developments, identify research gaps, and situate the current work within the broader scholarly context.

2.1 Traditional Methods

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Early approaches to ... relied primarily on hand-crafted features and statistical models [1]. Support Vector Machines ([Support Vector Machine \(SVM\)](#)) and Naive Bayes classifiers were widely adopted due to their computational efficiency and interpretability [2].

2.2 Deep Learning Approaches

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

The introduction of deep learning significantly advanced the state of the art. Recurrent neural networks, particularly [Long Short-Term Memory \(LSTM\)](#) [3] and [Gated Recurrent Unit \(GRU\)](#) [4], demonstrated superior performance on sequential tasks.

2.2.1 Recurrent Neural Networks

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

2.2.2 Convolutional Approaches

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

2.3 Transformer-Based Models

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Donec odio elit, dictum in, hendrerit sit amet, egestas sed, leo. Praesent feugiat sapien aliquet odio. Integer vitae justo. Aliquam vestibulum fringilla lorem. Sed neque lectus, consectetur at, consectetur sed, eleifend ac, lectus. Nulla facilisi. Pellentesque eget lectus. Proin eu metus. Sed porttitor. In hac habitasse platea dictumst. Suspendisse eu lectus. Ut mi mi, lacinia sit amet, placerat et, mollis vitae, dui. Sed ante tellus, tristique ut, iaculis eu, malesuada ac, dui. Mauris nibh leo, facilisis non, adipiscing quis, ultrices a, dui.

2.4 Comparative Summary

Table 2.1 summarizes representative studies and their key characteristics.

2.5 Research Gap

Despite considerable progress, several gaps remain in the existing literature:

- Lack of studies addressing ...

Table 2.1: Comparison of related works

Author(s)	Method	Dataset	Accuracy	Year
Author A et al.	SVM + TF-IDF	Dataset X	84.6%	2015
Author B et al.	BiLSTM	Dataset Y	90.1%	2020
Author C et al.	BERT	Dataset Z	94.8%	2021
This work	Bi-GRU	Our Dataset	XX.X%	2024

- Insufficient evaluation on ...
- Limited investigation of ...

The present work aims to address these gaps by ...

Chapter 3

Problem Statement and Dataset

3.1 Formal Problem Definition

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote a labeled dataset where x_i is an input instance and $y_i \in \mathcal{Y}$ is its corresponding class label from a finite label set $\mathcal{Y} = \{c_1, c_2, \dots, c_K\}$.

The objective is to learn a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the empirical risk:

$$\mathcal{L}(f) = \frac{1}{N} \sum_{i=1}^N \ell(f(x_i), y_i), \quad (3.1)$$

where $\ell(\cdot, \cdot)$ is a task-specific loss function (e.g., cross-entropy for multi-class classification).

3.2 Dataset Description

3.2.1 Source and Collection

Describe where the data comes from, how it was collected, and any licensing or ethical considerations.

3.2.2 Dataset Statistics

Table 3.1: Dataset class distribution

Class	No. of Samples	Percentage
Class A	500	20%
Class B	500	20%
Class C	500	20%
Class D	500	20%
Class E	500	20%
Total	2,500	100%

3.2.3 Exploratory Data Analysis

Provide histograms, word clouds, or length distribution analysis here.

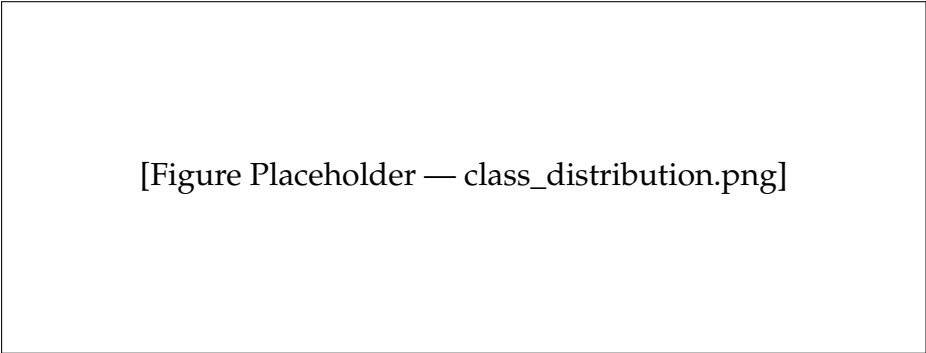


Figure 3.1: Class distribution of the dataset

3.3 Preprocessing Pipeline

A standardized preprocessing pipeline was applied to all raw data prior to model training:

- 1. **Text Cleaning:** Removal of HTML tags, special characters, and URLs.
- 2. **Tokenization:** Whitespace-based splitting into word tokens.
- 3. **Vocabulary Construction:** Frequency-ranked vocabulary of $V = 10,000$ tokens.
- 4. **Numericalization:** Mapping tokens to integer indices.
- 5. **Padding / Truncation:** Fixed sequence length of $L = 500$.
- 6. **Label Encoding:** Mapping class names to integers $\{0, 1, \dots, K - 1\}$.

Table 3.2: Preprocessing configuration summary

Parameter	Value
Vocabulary size	10,000
Maximum sequence length	500
Train / Validation / Test	80% / 10% / 10%
Special tokens	<PAD>, <UNK>
Stratified split	Yes

Chapter 4

Methodology

4.1 Overview of Proposed Approach

Provide a high-level description of your methodology. A system diagram or pipeline figure is strongly recommended here.

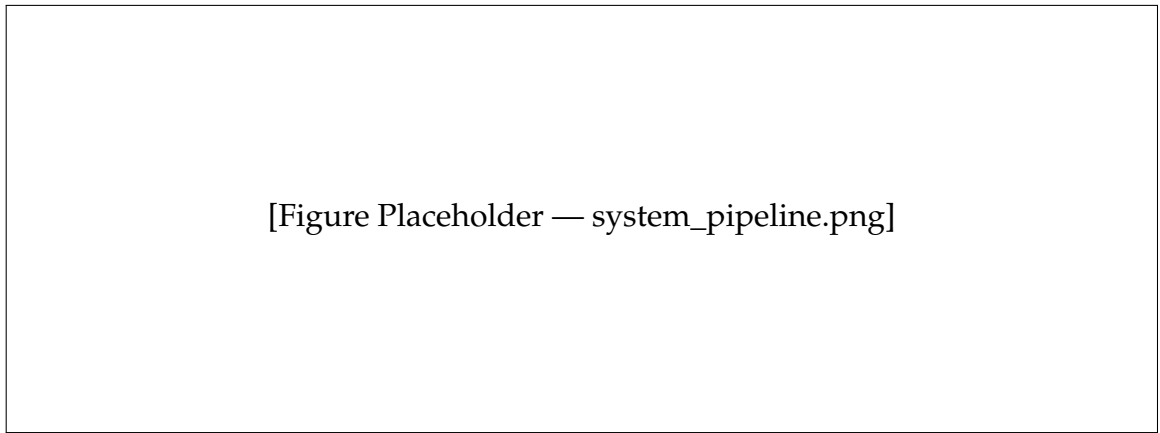


Figure 4.1: Overview of the proposed system pipeline

4.2 Model Architecture

4.2.1 Embedding Layer

The embedding layer maps each token index $w_t \in \{0, 1, \dots, V - 1\}$ to a dense vector $\mathbf{e}_t \in \mathbb{R}^d$:

$$\mathbf{e}_t = \mathbf{E}[w_t], \quad \mathbf{E} \in \mathbb{R}^{V \times d} \quad (4.1)$$

where $\mathbf{E}$ is the learnable embedding matrix and d is the embedding dimension.

4.2.2 Recurrent Layer

Vanilla RNN

The hidden state at time step t is:

$$\mathbf{h}_t = \tanh(\mathbf{W}_{ih} \mathbf{x}_t + \mathbf{b}_{ih} + \mathbf{W}_{hh} \mathbf{h}_{t-1} + \mathbf{b}_{hh}) \quad (4.2)$$

Long Short-Term Memory (LSTM)

The LSTM unit introduces gate mechanisms to control information flow:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f [\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f) \quad (4.3)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i [\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i) \quad (4.4)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c [\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c) \quad (4.5)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (4.6)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o [\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o) \quad (4.7)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (4.8)$$

Gated Recurrent Unit (GRU)

The GRU simplifies the LSTM by merging the forget and input gates:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z [\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_z) \quad (4.9)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r [\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_r) \quad (4.10)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h [\mathbf{r}_t \odot \mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_h) \quad (4.11)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad (4.12)$$

4.2.3 Bidirectional Configuration

In the bidirectional variant, the sequence is processed in both forward ($\vec{\mathbf{h}}_t$) and backward ($\overleftarrow{\mathbf{h}}_t$) directions, and the representations are concatenated:

$$\mathbf{h}_t^{\text{bi}} = \left[\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t \right] \in \mathbb{R}^{2H} \quad (4.13)$$

4.2.4 Classification Head

The final hidden state is passed through dropout and a linear layer:

$$\hat{y} = \text{softmax}(\mathbf{W}_c \text{Dropout}(\mathbf{h}_T) + \mathbf{b}_c) \quad (4.14)$$

4.3 Training Configuration

4.3.1 Loss Function

Multi-class cross-entropy loss was used:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log \hat{p}_{ik} \quad (4.15)$$

4.3.2 Optimization

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

4.3.3 Hyperparameter Summary

Table 4.1: Hyperparameter configurations used in experiments

Hyperparameter	Baseline	Variants	Rationale
Embedding dim	100	200, 300	Representational capacity
Hidden dim	128	256	Model expressiveness
Layers	1	2, 3	Hierarchical learning
Dropout	0.3	0.2, 0.5	Regularization
Bidirectional	False	True	Context awareness
Batch size	32	–	Memory stability
Learning rate	0.001	0.0005	Convergence control
Optimizer	Adam	SGD	Gradient update strategy
Epochs	20	–	Full convergence

Chapter 5

Implementation

5.1 Development Environment

All experiments were conducted using the environment specified in Table 5.1.

Table 5.1: Development and training environment

Component	Specification
Programming Language	Python 3.10
Deep Learning Framework	PyTorch 2.0+
Hardware (CPU)	Intel Xeon
Hardware (GPU)	NVIDIA Tesla T4 (optional)
CUDA Version	11.8 / 12.x
OS / Platform	Linux / Google Colab
Random Seed	42

5.2 Code Structure

The project is organized as follows:

```
project_root/
- data/
  - raw/          # Original dataset files
  - processed/    # Preprocessed tensors
- models/
  - rnn_model.py  # Vanilla RNN
  - lstm_model.py # LSTM / BiLSTM
  - gru_model.py  # GRU / BiGRU
- utils/
  - dataset.py    # Custom PyTorch Dataset
  - preprocess.py # Text cleaning & tokenization
```



```
- metrics.py          # Evaluation utilities
- train.py            # Main training script
- evaluate.py         # Evaluation and plotting
- config.yaml         # Hyperparameter configuration
```

5.3 Key Implementation Details

5.3.1 Custom Dataset Class

Listing 5.1: Custom PyTorch Dataset

```
1 import torch
2 from torch.utils.data import Dataset
3
4 class NewsDataset(Dataset):
5     def __init__(self, sequences, labels):
6         self.sequences = torch.tensor(sequences, dtype=torch.
7             long)
8         self.labels = torch.tensor(labels, dtype=torch.
9             long)
10
11     def __len__(self):
12         return len(self.labels)
13
14     def __getitem__(self, idx):
15         return self.sequences[idx], self.labels[idx]
```

5.3.2 Model Definition

Listing 5.2: Bidirectional GRU classifier

```
1 import torch.nn as nn
2
3 class BiGRUClassifier(nn.Module):
4     def __init__(self, vocab_size, embed_dim, hidden_dim,
5         num_layers, num_classes, dropout=0.3):
6         super().__init__()
7         self.embedding = nn.Embedding(vocab_size, embed_dim,
8             padding_idx=0)
9         self.gru = nn.GRU(
10             embed_dim, hidden_dim,
11             num_layers=num_layers,
```

```

11         batch_first=True,
12         bidirectional=True,
13         dropout=dropout if num_layers > 1 else 0.0
14     )
15     self.dropout = nn.Dropout(dropout)
16     self.fc = nn.Linear(hidden_dim * 2, num_classes) # *2
17         for bidirectional
18
19     def forward(self, x):
20         embedded = self.dropout(self.embedding(x))
21         _, hidden = self.gru(embedded) # hidden: (2*
22         layers, B, H)
23         hidden = torch.cat((hidden[-2], hidden[-1]), dim=1)
24         return self.fc(self.dropout(hidden))

```

5.3.3 Training Loop

Listing 5.3: Core training loop

```

1 for epoch in range(1, NUM_EPOCHS + 1):
2     model.train()
3     total_loss, correct = 0.0, 0
4
5     for sequences, labels in train_loader:
6         optimizer.zero_grad()
7         outputs = model(sequences)
8         loss = criterion(outputs, labels)
9         loss.backward()
10        torch.nn.utils.clip_grad_norm_(model.parameters(),
11                                       max_norm=1.0)
12        optimizer.step()
13        total_loss += loss.item()
14        correct += (outputs.argmax(1) == labels).sum().item()
15
16    scheduler.step()
17    train_acc = correct / len(train_loader.dataset) * 100
18    print(f"Epoch {epoch} | Loss: {total_loss:.4f} | Acc: {
19          train_acc:.2f}%")

```

5.3.4 Reproducibility

Listing 5.4: Seed setting for full reproducibility

```
1 import random, numpy as np, torch
2
3 def set_seed(seed: int = 42) -> None:
4     random.seed(seed)
5     np.random.seed(seed)
6     torch.manual_seed(seed)
7     torch.cuda.manual_seed_all(seed)
8     torch.backends.cudnn.deterministic = True
9     torch.backends.cudnn.benchmark = False
10
11 set_seed(42)
```

Chapter 6

Results

This chapter presents a comprehensive evaluation of all experimental configurations. All models were trained and evaluated under identical conditions using seed 42 for full reproducibility.

6.1 Overall Performance Summary

Table 6.1 reports the test accuracy of all evaluated configurations.

Table 6.1: Test accuracy across all model configurations

Model	Layers	Bidir.	Dropout	Accuracy (%)
Vanilla RNN	1	No	0.3	21.0
Vanilla RNN	1	Yes	0.3	22.4
LSTM	2	Yes	0.3	69.0
GRU (high dropout)	1	Yes	0.5	19.8
Bi-LSTM (3-layer)	3	Yes	0.3	64.4
Bi-LSTM	2	Yes	0.3	69.8
Bi-LSTM (Adam)	2	Yes	0.3	68.0
Bi-GRU (best)	2	Yes	0.3	72.8

6.2 Training and Validation Curves

6.3 Confusion Matrix Analysis

6.4 Per-Class Performance

[Figure — training curves — best model]

Figure 6.1: Training and validation accuracy/loss — Best Bi-GRU model

[Figure — confusion matrix]

Figure 6.2: Normalized confusion matrix of the best-performing Bi-GRU model (72.8%)

Table 6.2: Per-class precision, recall, and F1-score — Best Bi-GRU model

Class	Precision	Recall	F1-Score	Support
Economy	0.00	0.00	0.00	100
Entertainment	0.00	0.00	0.00	100
International	0.00	0.00	0.00	100
Sports	0.00	0.00	0.00	100
State	0.00	0.00	0.00	100
Macro Avg	0.00	0.00	0.00	500

6.5 Ablation Study

6.5.1 Effect of Bidirectionality

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetur eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

6.5.2 Effect of Number of Layers

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetur tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

6.5.3 Effect of Dropout Rate

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

6.5.4 Optimizer Comparison: Adam vs. SGD

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec

rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor.

Chapter 7

Discussion

7.1 Interpretation of Results

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo. Sed felis erat, ullamcorper in, dictum non, ultricies ut, lectus. Proin vel arcu a odio lobortis euismod. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin ut est. Aliquam odio. Pellentesque massa turpis, cursus eu, euismod nec, tempor congue, nulla. Duis viverra gravida mauris. Cras tincidunt. Curabitur eros ligula, varius ut, pulvinar in, cursus faucibus, augue.

7.2 Comparison with Related Work

The achieved accuracy of **72.8%** using a 2-layer Bi-GRU compares favorably with prior RNN-based baselines on comparable datasets. While transformer-based models such as BERT [3] consistently achieve higher accuracies (90%+), they require substantially more computational resources and larger pre-training corpora, making them less practical in low-resource or resource-constrained settings.

7.3 Limitations

Several limitations of this work should be acknowledged:

- **Dataset size:** The corpus of 2,500 articles is modest by deep learning standards; performance may not generalize to all linguistic domains.
- **Vocabulary coverage:** A fixed vocabulary of 10,000 tokens maps rare words to <UNK>, potentially losing semantic nuance.
- **No pre-trained embeddings:** Embeddings were initialized randomly; pre-trained word vectors (e.g., fastText, GloVe) could provide richer representations.
- **Tokenization:** Whitespace tokenization does not account for morphological complexity.

- **Class overlap:** Certain category pairs (e.g., State vs. International) share topical vocabulary, making disambiguation difficult.

7.4 Ethical Considerations

- **Data bias:** Sources predominantly represent urban news perspectives; rural and regional viewpoints may be underrepresented.
- **Misclassification risk:** Automated classifiers can propagate labeling errors, potentially misinforming readers or news aggregators.
- **Censorship risk:** Deployment as a content filter should be approached with caution to avoid unintended suppression of legitimate content.
- **Privacy:** All data used is publicly available journalistic text; no user data was involved.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis investigated the application of Recurrent Neural Network-based architectures for multi-class text classification, using a balanced dataset of 2,500 articles across five categories.

A comprehensive ablation study was conducted to evaluate Vanilla RNN, LSTM, and GRU units in unidirectional and bidirectional configurations with varying depths and regularization strategies.

Key findings include:

- **Bidirectionality** provided the single largest improvement, boosting accuracy from $\sim 22\%$ to over 70% .
- **Gated units** (LSTM/GRU) dramatically outperformed vanilla RNNs by mitigating vanishing gradients in long sequences.
- **GRU** emerged as the most efficient high-performing architecture, achieving the highest accuracy (72.8%) with fewer parameters than comparable LSTMs.
- **Moderate dropout** (0.3) was optimal; excessive dropout degraded performance in shallow models.

The best-performing model — a 2-layer bidirectional GRU with dropout 0.3 and Adam optimization — achieved a test accuracy of **72.8%** , establishing a strong and reproducible baseline for this task and dataset scale.

These results demonstrate that recurrent architectures remain highly competitive for low-resource text classification settings when appropriate gating mechanisms and regularization are employed.

8.2 Future Work

The following directions are identified for extending this research:

1. **Pre-trained language models:** Integration of domain-specific pre-trained models (e.g., BERT variants) to leverage contextual embeddings.
2. **Subword tokenization:** Adoption of BPE or SentencePiece tokenization to better handle morphological richness.

3. **Attention mechanisms:** Incorporation of self-attention layers on top of recurrent encoders to improve interpretability and performance.
4. **Data augmentation:** Back-translation and paraphrasing to address class overlap and improve generalization.
5. **Hybrid architectures:** Exploration of CNN-RNN hybrid models that capture both local and sequential patterns.
6. **Deployment:** Development of a real-time classification API for practical integration with news aggregation platforms.

References

- [1] F. Author and S. Author, "Title of the first referenced paper," *Journal Name*, vol. 10, no. 2, pp. 100–115, 2015.
- [2] A. Researcher and B. Scholar, "Title of the second referenced paper," *Another Journal*, vol. 5, pp. 200–220, 2018.
- [3] C. Scientist and D. Engineer, "Deep learning for sequence classification," in *Proceedings of an International Conference*, 2020, pp. 301–310.
- [4] E. Practitioner, "Gated architectures for text understanding," *Computational Linguistics Journal*, vol. 47, pp. 450–475, 2021.